

Deducing probabilistic security feature expressions from a SVO model

Fernando Román-García; Juan Hernández-Serrano; Oscar Esparza

1 Introduction

In this document, we rely on the formal security analysis conducted in the paper that introduces the NoRDEx protocol. Specifically, we use this analysis to derive the probabilistic expressions defined in the same paper to describe the security features.

We first present the goals and premises used in the SVO model (see Section 2). Then, we describe the process to obtain the security features (see Section 3).

2 SVO model

The complete model is defined in the NoRDEx paper. However, below we present the goals of the security analysis:

- G1.** $\mathcal{J}$ believes ($\mathcal{P}$ said c)
- G2.** $\mathcal{J}$ believes ($\mathcal{C}$ received c)
- G3.** $\mathcal{J}$ believes ($\mathcal{DLT}$ said k)
- G4.** $\mathcal{J}$ believes ($\mathcal{P}$ said m)
- G5.** $\mathcal{J}$ believes ($\mathcal{C}$ received m)

And next, we present the premises:

- P1.** $\mathcal{J}$ received (c, π_{PoO})
- P2.** $\mathcal{J}$ received (c, π_{PoI})
- P3.** $\mathcal{J}$ believes $V_{Agr}(a, \mathcal{C}, \mathcal{P})$
- P4.** $\mathcal{J}$ believes $V_{PoO}(\pi_{PoO}, a)$
- P5.** $\mathcal{J}$ believes $V_{PoI}(\pi_{PoI}, a, \emptyset)$
- P6.** $\mathcal{J}$ believes $V_{Ciph}(c, \pi_{PoO})$
- P7.** $\mathcal{J}$ believes $V_{PoO}(\pi_{PoO}, a)$
 $\quad \subset \mathcal{J}$ believes $V_{Sig}(\pi_{PoO}, pub_{\mathcal{P}})$
- P8.** $\mathcal{J}$ believes $V_{PoI}(\pi_{PoI}, a, \emptyset)$
 $\quad \subset \mathcal{J}$ believes $V_{Sig}(\pi_{PoI}, pub_{\mathcal{C}})$
- P9.** ($\mathcal{J}$ received $a \wedge \mathcal{J}$ believes $V_{Agr}(a, \mathcal{C}, \mathcal{P})$)
 $\quad \supset \mathcal{J}$ believes $PK_{\sigma}(\mathcal{P}, pub_{\mathcal{P}})$
- P10.** ($\mathcal{J}$ received $a \wedge \mathcal{J}$ believes $V_{Agr}(a, \mathcal{C}, \mathcal{P})$)
 $\quad \supset \mathcal{J}$ believes $PK_{\sigma}(\mathcal{C}, pub_{\mathcal{C}})$
- P11.** ($\mathcal{P}$ said $(\pi_{PoO}.type) \wedge \mathcal{J}$ believes $V_{PoO}(\pi_{PoO}, a)$
 $\quad \wedge \mathcal{J}$ received $c \wedge \mathcal{J}$ believes $V_{Ciph}(c, \pi_{PoO})$)
 $\quad \supset \mathcal{J}$ believes ($\mathcal{P}$ said c)
- P12.** ($\mathcal{C}$ said $(\pi_{PoI}.type) \wedge \mathcal{J}$ believes $V_{PoI}(\pi_{PoI}, a, \emptyset)$
 $\quad \wedge \mathcal{J}$ received $c \wedge \mathcal{J}$ believes $V_{Ciph}(c, \pi_{PoO})$)
 $\quad \supset \mathcal{J}$ believes ($\mathcal{C}$ received c)
- P13.** $\mathcal{J}$ received $\pi_{PoK} \wedge \mathcal{J}$ believes $V_{PoK}(\pi_{PoK}, a, \emptyset)$
- P14.** $\mathcal{J}$ believes $V_{Key}(k, t_k, \pi_{PoO})$
- P15.** $\mathcal{J}$ believes $V_{Sch}(m, a.sch)$
- P16.** $\mathcal{J}$ believes ($D(c, k) \neq \perp$)

P17. ($\mathcal{J}$ believes ($\mathcal{P}$ said c)
 $\wedge \mathcal{J}$ believes ($\mathcal{DLT}$ said $k \vee V_{PoK}(\pi_{PoK}, a, \emptyset)$)
 $\wedge \mathcal{J}$ believes $V_{Key}(k, t_k, \pi_{PoO})$
 $\wedge \mathcal{J}$ believes $V_{Sch}(m, a.sch)$
 $\wedge \mathcal{J}$ believes ($D(c, k) \neq \perp$)
 $\supset \mathcal{J}$ believes ($\mathcal{P}$ said m))

P18. ($\mathcal{J}$ believes ($\mathcal{C}$ received c)
 $\wedge \mathcal{J}$ believes ($\mathcal{DLT}$ said k)
 $\wedge \mathcal{J}$ believes $V_{Key}(k, t_k, \pi_{PoO})$
 $\wedge \mathcal{J}$ believes $V_{Sch}(m, a.sch)$
 $\wedge \mathcal{J}$ believes ($D(c, k) \neq \perp$)
 $\supset \mathcal{J}$ believes ($\mathcal{C}$ received m))

3 Security feature deduction

In this section, we will use the conclusions obtained by applying the SVO model to NoRDEx to build expressions that allow us to easily understand the necessary conditions for the non-repudiation of origin (NRO) security feature to hold. We only detail the deduction process for one of them, since the processes are very similar and describing both would unnecessarily complicate the explanation.

3.1 From logic to probabilistic expressions

To do so, we start from the equation used for Demonstration 3 (see NoRDEx paper). In Equation 1, we can quickly identify that, for a judge to believe that the provider sent the message (G4), a series of conditions must hold ($G1 \wedge (G3 \vee P13) \wedge P14 \wedge P15 \wedge P16$). Premise P17 is the most critical one, as it defines what we consider the act of sending the message (it links the conditions to the event we want to prove).

$$\frac{(G1 \wedge (G3 \vee P13) \wedge P14 \wedge P15 \wedge P16) \wedge (P17)}{G4} \quad (1)$$

Since G4 is an event, and we have identified a set of conditions that must always hold for it to occur, we can translate the logical expression into a probabilistic formula:

$$\Pr \left[G4 \mid \begin{array}{l} G1 \wedge (G3 \vee P13) \wedge \\ P14 \wedge P15 \wedge P16 \end{array} \right] = 1 \quad (2)$$

3.2 Translating SVO to the protocols' notation

Next, we need to transform Equation 2 to make it easier to read, using the notation we defined for the protocol description. First, we need to determine what G4 means. We consider it reasonable to state that the message could only have been sent if the NoRDEx protocol was executed – at least the first and third phases – since these are the phases where the consumer obtains the necessary evidence to present to a judge (Equation 3).

$$G4 \equiv \left(\begin{array}{l} \mathbf{Exc}_{PoO} \langle \mathcal{C}_0, \mathcal{P}_0 \rangle \wedge \\ \mathbf{Exc}_{PoK} \langle \mathcal{C}_2, \mathcal{P}_2, \mathcal{DLT}() \rangle \end{array} \right) \quad (3)$$

By substituting Equation 3 into Equation 2, we obtain:

$$\Pr \left[\begin{array}{l} \mathbf{Exc}_{PoO} \langle \mathcal{C}_0, \mathcal{P}_0 \rangle \wedge \\ \mathbf{Exc}_{PoK} \langle \mathcal{C}_2, \mathcal{P}_2, \mathcal{DLT}() \rangle \end{array} \mid \begin{array}{l} G1 \wedge (G3 \vee P13) \wedge \\ P14 \wedge P15 \wedge P16 \end{array} \right] = 1 \quad (4)$$

Next, we need to understand what G1 implies, so we use Demonstration 3 (see NoRDEx paper), from which we can derive the following formula:

$$\Pr [G1 \mid P1 \wedge P3 \wedge P4 \wedge P6] = 1 \quad (5)$$

En Equation 6 cambiamos las expresiones de SVO por expresiones que únicamente utilizan la notación con la que hemos descrito el protocolo.

$$\begin{aligned}
P1 &\equiv \mathcal{C}(c, \pi_{PoO}) \\
P3 &\equiv V_{\text{Agr}}(a, \mathcal{C}, \mathcal{P}) \\
P4 &\equiv V_{\text{PoO}}(\pi_{PoO}, a) \\
P6 &\equiv V_{\text{Ciph}}(c, \pi_{PoO})
\end{aligned} \tag{6}$$

Definition 3.1 *If we know that $\Pr[A \mid B] = 1$ and $\Pr[B \mid C] = 1$, then we know that $\Pr[A \mid C] = 1$.*

By applying Definition 3.1 and Equations 4 and 6, we obtain the following expression:

$$\Pr \left[\begin{array}{c} \mathbf{Exc}_{PoO}(\mathcal{C}_0, \mathcal{P}_0) \wedge \\ \mathbf{Exc}_{PoK}(\mathcal{C}_2, \mathcal{P}_2, \mathcal{DLT}()) \end{array} \middle| \begin{array}{c} \mathcal{C}(c, \pi_{PoO}) \wedge v_1 \\ (G3 \vee P13) \wedge \\ P14 \wedge P15 \wedge P16 \end{array} \right] = 1 \tag{7}$$

$$\text{where } \left\{ \begin{array}{l} v_1 \leftarrow \left(\begin{array}{l} V_{\text{Agr}}(a, \mathcal{C}, \mathcal{P}) \wedge \\ V_{\text{PoO}}(\pi_{PoO}, a) \wedge \\ V_{\text{Ciph}}(c, \pi_{PoO}) \end{array} \right) \end{array} \right.$$

The expressions G3 and P13 can also be understood as:

$$\begin{aligned}
G3 &\equiv \mathcal{DLT}((eid, k, t_k)) \\
P13 &\equiv \mathcal{C}(\pi_{PoK}) \wedge V_{\text{PoK}}(\pi_{PoK}, a, \emptyset)
\end{aligned} \tag{8}$$

If we apply Definition 3.1 to Equations 7 and 8, we obtain:

$$\Pr \left[\begin{array}{c} \mathbf{Exc}_{PoO}(\mathcal{C}_0, \mathcal{P}_0) \wedge \\ \mathbf{Exc}_{PoK}(\mathcal{C}_2, \mathcal{P}_2, \mathcal{DLT}()) \end{array} \middle| \begin{array}{c} \mathcal{C}(c, \pi_{PoO}, \pi_{PoK}) \wedge \\ (v_1 \wedge v_2) \wedge \\ P14 \wedge P15 \wedge P16 \end{array} \right] = 1 \tag{9}$$

$$\text{where } \left\{ \begin{array}{l} v_1 \leftarrow (\mathcal{DLT}((eid, k, t_k)) \vee V_{\text{PoK}}(\pi_{PoK}, a, \emptyset)) \\ v_2 \leftarrow \left(\begin{array}{l} V_{\text{Agr}}(a, \mathcal{C}, \mathcal{P}) \wedge \\ V_{\text{PoO}}(\pi_{PoO}, a) \wedge \\ V_{\text{Ciph}}(c, \pi_{PoO}) \end{array} \right) \end{array} \right.$$

Note that we have grouped $\mathcal{C}(c, \pi_{PoO})$ and $\mathcal{C}(\pi_{PoK})$ into $\mathcal{C}(c, \pi_{PoO}, \pi_{PoK})$. Now, we need to transform P14, P15, and P16:

$$\begin{aligned}
P14 &\equiv V_{\text{Key}}(k, t_k, \pi_{PoO}) \\
P15 &\equiv V_{\text{Sch}}(m, a, sch) \\
P16 &\equiv (D(c, k) \neq \perp)
\end{aligned} \tag{10}$$

Finally, by combining Equations 9 and 10 using Definition 3.1, we obtain Equation 11. For completeness, we have included the exact definition of the variables $\mathcal{C}_0$, $\mathcal{C}_2$, $\mathcal{P}_0$, and $\mathcal{P}_2$. The advantage of using this expression is that it allows us to easily identify the necessary conditions for the **NRO security feature of m** to hold.

$$\begin{aligned}
& \Pr \left[\begin{array}{c} \mathbf{Exc}_{PoO}(\mathcal{C}_0, \mathcal{P}_0) \wedge \\ \mathbf{Exc}_{PoK}(\mathcal{C}_2, \mathcal{P}_2, \mathcal{DLT}()) \end{array} \middle| \begin{array}{c} \mathcal{C}(c, \pi_{PoO}, \pi_{PoK}) \wedge \\ (v_1 \wedge v_2) \end{array} \right] = 1 \\
& \text{where } \left\{ \begin{array}{l} v_1 \leftarrow (\mathcal{DLT}((eid, k, t_k)) \vee V_{PoK}(\pi_{PoK}, a, \emptyset)) \\ v_2 \leftarrow \left(\begin{array}{l} V_{Agr}(a, \mathcal{C}, \mathcal{P}) \wedge \\ V_{PoO}(\pi_{PoO}, a) \wedge \\ V_{Ciph}(c, \pi_{PoO}) \wedge \\ V_{Key}(k, t_k, \pi_{PoO}) \wedge \\ V_{Sch}(m, a.sch) \wedge \\ (D(c, k) \neq \perp) \end{array} \right) \\ \mathcal{C}_0 \leftarrow \mathcal{C}(a, path) \\ \mathcal{P}_0 \leftarrow \mathcal{P}(a, priv_{\mathcal{P}}) \\ \mathcal{C}_2 \leftarrow \mathcal{C}(a, c, \pi_{PoO}) \\ \mathcal{P}_2 \leftarrow \mathcal{P}(k, \pi_{PoI}, priv_{\mathcal{P}}) \end{array} \right. \quad (11)
\end{aligned}$$